Mathematical overview

Copies alone preserve the all-red and all-blue states, so those two states can never merge. Each copy leaves every coordinate dependent on one original coordinate. A color reset creates a constant coordinate. Exactly one reset followed by two appropriate copies synchronizes all eight three-slot starts, and three instructions is optimal: with fewer, at least one of the three coordinates has never been a destination and still retains its initial color. This universal claim is stronger than repairing a chosen pair.

For independent uniform draws of position tickets with replacement, coverage occurs in 6 of the 27 length-three stories and 36 of the 81 length-four stories. Each story of a given length has equal probability. No finite deadline guarantees coverage: a story repeating ticket 1 leaves two positions unseen. The probability of never covering is zero, since the probability a particular position is missing after m draws is (2/3)^m, and the chance at least one is missing is at most 3(2/3)^m. This is an eventual probabilistic statement, not an observed classroom result. Without replacement, three draws cover all positions deterministically.

For S=(set left to red) and T=(cyclic left rotation), STSTS makes every start RRR in five instructions. Three S instructions are necessary because only S can remove one surviving original dependency; two T instructions are necessary to bring all three original slots to the left. Thus five is minimal. The positions may move: counting physical slots visited by a reset is insufficient for this device.

Children discover concrete obstructions and design universal words (2-5), then distinguish exact finite history counts from a random deadline (4-5). Trials suggest a claim. The monochromatic counterexample, dependency argument, and exhaustive history counts establish the printed conclusions. The familiar base rules are not automatically valid after changing to copies or rotation.

Entry route, readiness, preparation, and launch

Unpiloted; exact counter fit, ticket handling, and rotating a row have not been physically rehearsed. Page 1 is an accessible entry for children who follow left-to-right positions and can copy a color. Reading can be supplied by an adult. The universal/shortest questions suit grades 2-5, by readiness. Page 2 needs counting ordered three- and four-draw stories, multiplication through 81, and the distinction between possible and guaranteed. Fraction notation is optional. Page 3 needs retaining a moving row and recording a word. K-1 may help operate boards but is not assigned an independent version of the whole investigation.

Per pair: two printed three-slot working rows, six reversible red/blue counters, three same-size tickets 1,2,3, a cup, three small coverage markers, pencil and scrap paper. If counters are not reversible, provide six red and six blue paper circles. Large circles are 21.9 mm across; use counters no wider than 20 mm. The example diagrams are smaller and are not working mats.

For the eleven-child KK11 / 3333 / 445 arrangement: five pair kits (two at K-1, two at 2-3, one at 4-5); the third older child checks instructions and rotates roles. Totals: ten working rows, thirty reversible counters (or thirty of each single color), fifteen tickets, five cups, fifteen coverage markers, five pencils, and three anchored adults. K-1 can return to base overwrite work or join the concrete copy experiment with adult support; page 2 is primarily an older-table return visit.

Start with 3-4 minutes of handling the counters. Demonstrate COPY 1 to 2 on the exact paired visual: RBR becomes RRR; BRB becomes BBB. Both boards use the same instruction but copy their own current color. Ask for a legal next instruction. Let the partner check each move. Introduce S/T only on a return visit or once copies are familiar: physically pick up all three counters together and rotate, rather than changing their colors.

Suggested route: 12-20 minutes on page 1, 15-25 minutes on page 3, and a separate 20-35 minute return for page 2. There is no obligation to finish these in one meeting. Keep one recoverable instruction word and the final board; derive counts from the word later. Save discoveries and the last unfinished question in the use log.

Precise solutions, hints, and extensions

1. Any two starts that differ in every slot can never be made alike by copies: RRR and BBB, or the pictured RBR and BRB. A copy story gives each slot the starting color of some slot, the same slot on both boards, and the starts differ there. Starts that agree in some slot can be made alike by copying from it. RESET 1 to R; COPY 1 to 2; COPY 1 to 3 sends every start to RRR. A story with at most two instructions cannot target all three destination slots, so a starting difference at an untouched slot survives. This proof concerns a fixed story, independent of the starts. Optional hint: place a letter x, y, or z under each original counter, move that letter with a copy, and erase it only when reset. Extensions: constrain which directed copies are allowed; synchronization from one reset is possible exactly when the reset site can reach every site in that directed graph, if repeated copies are allowed. Test small graphs before generalizing.

2. Three draws: all three labels must appear once, so 3 x 2 x 1 = 6. Four draws: one label appears twice and the other two once. Choose the repeated label (3), its two positions (6), and the order of the remaining labels (2): 36. Thus probabilities are 2/9 and 4/9. The 81 stories may be pooled across pairs; children need not transcribe every story. Without replacement the six possible length-three stories all cover. With replacement, 111...1 is a counterexample for every chosen finite deadline. Optional hint: first sort covering four-draw stories by the repeated ticket. Extension: if each visited position also receives an independent fair new color, then conditional on a covering position story the final three colors are independent fair colors (use last visits), even without replacement. This does not hold for an arbitrary stopping rule inspecting colors.

3. From (x,y,z), STSTS gives (R,y,z), (y,z,R), (R,z,R), (z,R,R), (R,R,R). Neither rotation nor a reset at an already constant dependency removes a new original color. Each S removes at most one, requiring at least three S's. With fewer than two T's at most two original counters can have visited the reset position. The displayed five-letter word attains both bounds. Optional hint: move three differently marked counters and watch which original ones remain. Extension: n slots require 2n-1 operations in the same left-rotation/left-reset model, using S(TS)^(n-1); argue that n resets and at least n-1 rotations are needed. A requirement to restore the original orientation changes the optimum and is not imposed here.

New versus base, sources, and checks

Base Week 46 uses direct coordinate assignments, toggles, chosen coverage words, and last assignments. Page 1 introduces copying/dependency transport; page 2 counts waiting histories and changes replacement; page 3 changes the memory device to moving slots with a single reset location. These are three investigations within one theme, not three new themes.

Original extensions, not copied from a downloaded problem. Base source: Gautam Iyer's CMU probability notes, Coupling, subsection “Lazy random walk on the binary hypercube,” Proposition 9 proof sketch, as identified in the base adult guide. The choice to offer separate return visits follows the local Week 1 encore model and the coordinator's reading of Math Circle by the Bay, preface printed pp. viii-x (PDF pp. 9-11); that passage supports pedagogy, not these specific problems.

student/verify.py enumerates every three- and four-draw position history, all words over S/T through length five on all eight starts, and the one-reset/two-copy construction on all eight starts. It finds 6/27, 36/81, and the unique shortest word STSTS. The general impossibility/optimality arguments above are separate from those finite checks. No classroom or physical-pretest claim is made.